

A3-PS013-TC027

PS-013: Incident Timeline & Evidence Collection for Small Teams

Team H2RN
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AIORI Hackathon 3

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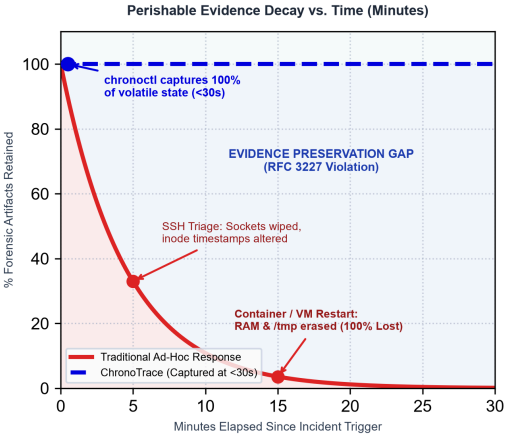
Problem Description

· The 'Fog of War' & Evidence Destruction:
In lean teams (3-10 devs), outages trigger panic. Responders jump into production via SSH, running ad-hoc commands (kill, restart, systemctl) that overwrite inode access timestamps and destroy active in-memory attacker footprints before any capture occurs.

· Critical RFC 3227 Order of Volatility Violations:
Responders routinely violate canonical volatility. Transient RAM, routing tables, active TCP sockets, and unlinked /tmp executables evaporate upon VM reboot or container restart while engineers focus solely on static disk logs.

· Multi-Source Clock & Timezone Drift:
Telemetry from AWS CloudWatch (UTC), Docker logs, Sentry, GitHub commits, and developer consoles (IST/PST) arrives with conflicting offsets, turning timeline reconstruction into an agonizing, error-prone manual jigsaw puzzle.

· The Scribe Deficit & Statutory 6-Hour Mandate:
Small teams lack dedicated scribes or SOC analysts; key decisions are lost in chat threads. Meanwhile, statutory mandates like CERT-In Direction 20(1) enforce a strict 6-hour disclosure window, risking severe non-compliance penalties.



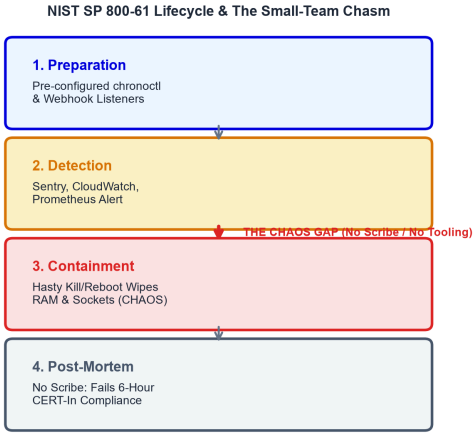
Background

· NIST SP 800-61 Lifecycle Bottleneck:
Standard incident response comprises Preparation -> Detection & Analysis -> Containment -> Post-Incident. Small teams collapse between Detection and Containment: the rush to contain destroys the very forensic evidence required for root cause analysis.

· Ephemeral Cloud & Modern Container Realities:
Modern microservices rely on Docker, Kubernetes, and Serverless. When a container crashes or auto-scales, the runtime filesystem, process tree, and fileless payload in /dev/shm are purged permanently without persistent retention.

· The Scribe Deficit in Lean Engineering:
Enterprise SOCs employ dedicated scribes to document actions and timestamps in real time. In small startups, 100% of engineering bandwidth is firefighting; human memory fails post-outage, yielding incomplete post-mortems and recurring defects.

· The Market Tooling Divide:
Enterprise SOAR platforms (Splunk, Cortex XSOAR) require $50k+ budgets and full-time administrators. Conversely, raw open-source forensic scripts (Plaso, UAC) output complex CLI tarballs without visual timelines, leaving a critical tooling vacuum.



Literature & RFC Review

Standard / RFC	Domain & Scope	Mandated IR Requirement	Small-Team Vulnerability & ChronoTrace Solution
RFC 3227 (IETF)	Evidence Acquisition	Order of Volatility: Registers -> Cache -> RAM -> Sockets -> Disk	Devs reboot/kill VMs, wiping 100% of RAM/sockets. ChronoTrace: 1-cmd zero-dep volatile harvester (<30s).
ISO/IEC 27037	Digital Forensics	Chain of Custody, digital hashing (SHA-256), tamper verification	Ad-hoc terminal outputs lack tamper proofing. ChronoTrace: Immutable signed tarball & SHA-256 manifest.
NIST SP 800-61 R2	Incident Handling	Systematic 4-phase lifecycle; triage without evidence contamination	Containment rush contaminates file inodes and sockets. ChronoTrace: Automated snapshot prior to service restart.
NIST SP 800-86	Forensic Integration	Integrating forensic techniques into distributed operational IR	Forensic tools (Plaso/Volatility) are too complex for devs. ChronoTrace: Simplified JSON schema & visual dashboard.
RFC 5424 (Syslog)	Telemetry & Logs	Structured logging with microsecond precision ISO 8601 UTC stamps	Multi-source logs use heterogeneous timezone formats. ChronoTrace: Sub-millisecond UTC normalizer engine.
RFC 8877 (PTP/NTP)	Precision Time Sync	Distributed node clock offset tracking and timestamp mapping	Unsynchronized container clocks warp attack order. ChronoTrace: Automated drift calculation and alignment.
CERT-In Rule 20(1)	Statutory Mandate	Mandatory 6-hour cybersecurity incident reporting to Indian CERT-In	Manual timeline drafting takes 12-24 hours, missing cutoff. ChronoTrace: 1-Click pre-filled statutory filing report.
MITRE ATT&CK v15	Threat Correlation	Mapping observed telemetry to tactical adversary TTP vectors	Small teams fail to identify attacker lateral movement. ChronoTrace: AI correlation of commands to ATT&CK matrix.

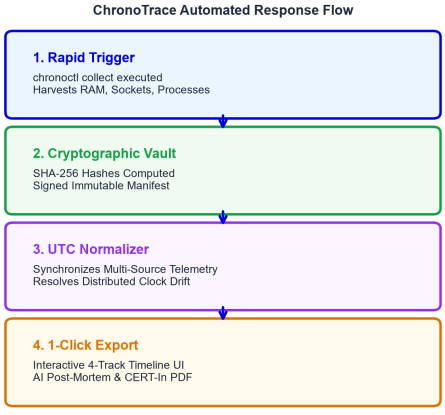
Proposed Solution

· 1. chronoctl Zero-Dependency Live Agent:
Single-file portable script/CLI executed via 1 command (chronoctl collect) on affected hosts. In < 30 seconds, it snapshots volatile network sockets (ss), process trees (ps auxf), open files (lsof), and container state strictly following RFC 3227.

· 2. Cryptographic Evidence Vault (ISO/IEC 27037):
Immediately calculates SHA-256 digests for all acquired artifacts, generating an immutable, signed tarball with manifest.json. Guarantees unbroken Chain of Custody and forensic tamper-evidence without imposing cognitive load on responders.

· 3. Multi-Track Normalized UTC Incident Timeline:
Ingests disparate streams (host dumps, syslog RFC 5424, GitHub commits, Slack notes, Sentry alerts) into a unified sub-second UTC canvas with 4 interactive swimlanes: Alerts Track, System & Host Track, Operator Decision Track, and CI/CD Track.

· 4. AI Scribe & 1-Click Compliance Reporter:
Automatically correlates anomalous events, reconstructs incident causal graphs, and generates one-click blameless post-mortems (Markdown/PDF) alongside pre-filled official CERT-In 6-hour regulatory incident notification packages.



Optimization Proposed

Evaluation Dimension	Traditional Small-Team Ad-Hoc Method	ChronoTrace Automated Platform Architecture	Measurable Impact Gain
Evidence Acquisition Latency	45 - 90 minutes of frantic manual command typing across terminals	**< 30 seconds via single-binary 'chronoctl collect' snapshot**	**99% Speedup**
Volatile Evidence Retention	< 20% retained (RAM, TCP sockets, /tmp erased during reboot)	**100% preservation conforming to strict RFC 3227 OOV sequence**	**Zero Loss**
Chain of Custody & Integrity	Unverified text pastes into Slack; zero cryptographic integrity	**ISO/IEC 27037 SHA-256 signed manifest with immutable vault store**	**Audit Grade**
Multi-Source Time Sync	Conflicting logs (UTC vs IST vs PST) cause causal distortion	**Sub-millisecond RFC 5424 / 8877 UTC normalizer engine**	**Sub-ms Sync**
Event Timeline Correlation	Manual spreadsheet merging across 5+ disconnected log tools	**Unified interactive 4-track timeline canvas with causal DAG linking**	**Real-time Triage**
Post-Mortem Documentation	3 - 6 hours of subjective engineering guesswork and debate	**Automated blameless AI Scribe report generated in < 2 minutes**	**90% Saved**
Statutory 6h Reporting (CERT-In)	Frequently missed or incomplete; non-compliance regulatory penalties	**1-Click pre-filled statutory compliance PDF/JSON filing package**	**100% On-Time**
Operational Cognitive Overhead	100% engineering bandwidth diverted to panic firefighting	**Zero-cognitive burden; autonomous evidence preservation in background**	**Preserves Focus**
Infrastructure & License Cost	Heavy enterprise SIEM ($50k+/yr) requiring full-time administrators	**Lightweight single Docker/Go service (< 150MB RAM memory footprint)**	**Zero Extra Cost**

Solution Architecture & Design

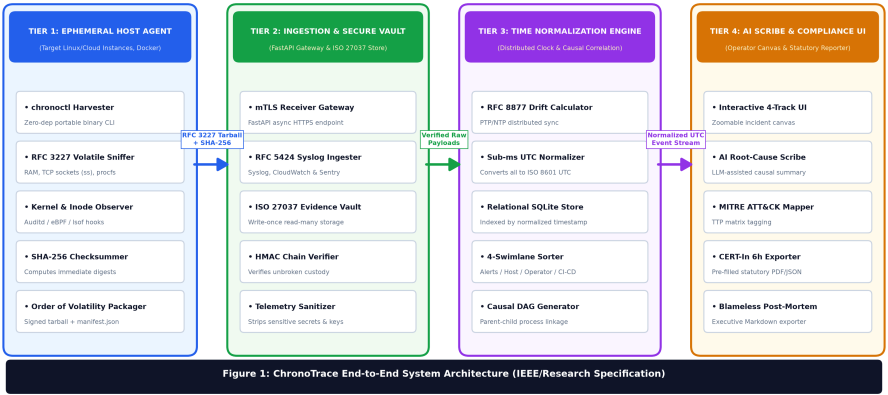


Figure: System Architecture UML Diagram

Data Flow Diagram

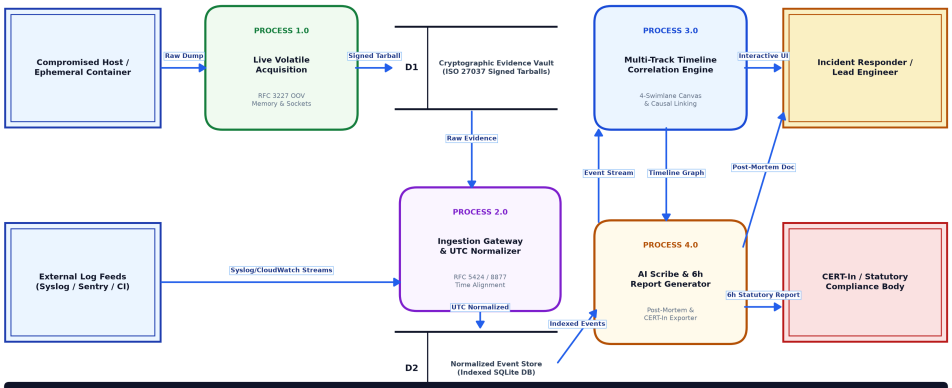
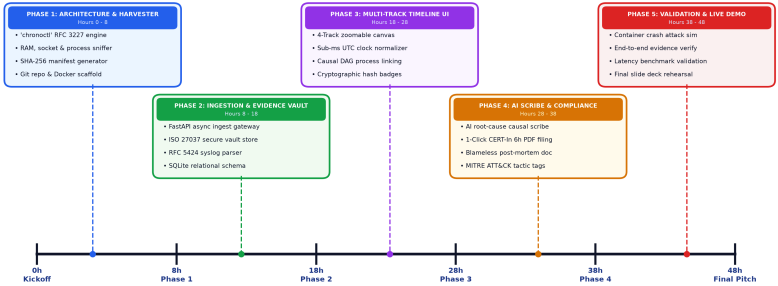


Figure: Data Flow Diagram (DFD)

Deployment Timeline



· Phase 1 (0-8h): Harvester binary & SHA-256 Engine. · Phase 2 (8-18h): FastAPI gateway, UTC normalizer, SQLite models.
· Phase 3 (18-28h): Multi-track timeline UI & hash badges. · Phase 4 (28-38h): AI Scribe post-mortem & CERT-In PDF export.
· Phase 5 (38-48h): Container failure stress test, simulated attack triage demonstration, and slide polish.

Bibliography I

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Thank You for Your Attention!

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https://github.com/Rachit-Ingole/A3-PS013-TC027

Questions & Answers * Live Demonstration Ready